

The plan, explained from zero: what we know, what we do next, and why — pen-and-paper edition

Cowork note for Chris — HardDisks / hspist3 (companion to the hand-calculation note)

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How to use this

Read it with the hand-calculation note next to you: that one has the derivations, this one has the *story* — what each result means, what we do next and what number we expect. Every step has a box “numbers to write down” (copy them into your notes), and many have “what a wrong result looks like” so you can judge Claude Code’s reports yourself. Exercises are scattered through the text; answers are in Section 5.

Units: $k_B T = m = \sigma = 1$, $\sigma = 2r$ is the disk diameter, speeds in $\sqrt{k_B T/m}$, times in $\sigma \sqrt{m/k_B T}$ (“ σ -time”; pixel time /24).

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1 The two papers in one picture

In plain words

Paper 1 (equilibrium). “Our simulator is a correct hard-disk gas.” We prove it by measuring two things that theory predicts to a fraction of a percent: the pressure $Z(\eta)$ and the speed of sound $c_s(\eta)$. If both match the Kolafa–Rottner equation of state from the ideal gas up to the dense fluid, the machine is trustworthy.

Paper 2 (non-equilibrium). “When we push this gas with a piston, the energy we lose and the information we lose obey the known laws.” Every law we want to test (energy conservation, Jarzynski, linear response, the nostalgia bound) is a statement about the *same* gas — so Paper 2 stands on Paper 1. If the gas were slightly wrong, every Paper 2 number would be slightly wrong and we would never know which law failed.

The chain, top to bottom:

	what we measure	against what
Paper 1	$Z(\eta)$ and $c_s(\eta)$, bulk values	Kolafa–Rottner EOS; $\sqrt{2}$ at $\eta \rightarrow 0$
Level 0	energy and momentum book-keeping of one push	must close to round-off
Level 1	divider jiggling in equilibrium	equipartition; gives the friction ζ
Level 2	work distribution $P(W)$ for many pushes	slow limit, fast limit, Jarzynski
Level 3	dissipation vs piston speed	linear at small speed, slope ζ
Level 4	memory, prediction, nostalgia	≥ 0 ; bound by dissipation

2 What we know today, result by result

2.1 Pressure: validated up to $\eta = 0.65$

In plain words

Pressure is force per length on the walls. There are two ways to get it out of a hard-disk run: count how hard the disks hit the *walls* (Z_{wall}), or count how hard they hit *each other* (Z_{pair} , the virial). Momentum conservation says the two must agree in a steady state — that is a built-in self-check. Then there is the box-size problem: a box with 100 disks is not an infinite fluid. The correction shrinks like $1/\sqrt{N}$, so we ran $N = 100, 400, 900, 1600$, plotted Z against $1/\sqrt{N}$, and read the value at $1/\sqrt{N} = 0$ (an infinite box).

$$Z(N) = Z_\infty + \frac{a}{\sqrt{N}}, \quad \chi^2 = \sum_N \frac{[Z_N - Z_\infty - a/\sqrt{N}]^2}{\sigma_N^2} \quad (1)$$

(4 points, 2 parameters $\Rightarrow$ 2 degrees of freedom).

What chi-squared means

If the four points really lie on a straight line and the error bars are honest, χ^2 comes out around its number of degrees of freedom (here ≈ 2). $\chi^2 = 0.07$ means “perfectly on a line” (maybe error bars a bit large). $\chi^2 = 7.6$ with 1 dof has probability 0.6% under the line hypothesis: the points do *not* lie on a $1/\sqrt{N}$ line, so no bulk value can be read off. That is what happens at $\eta = 0.67$ — not a bug, the fluid is starting to change structure there.

Numbers to write down

- Z_∞ within $\pm 0.17\%$ of KR for $\eta = 0.2\text{--}0.5$; $+0.13\%$ at 0.60; $-0.42 \pm 0.09\%$ at 0.65.
- $\eta = 0.67$: $\chi_1^2 = 7.6$ ($p = 0.006$); $\eta = 0.69$: slope $a = 8.7$ — no bulk value. $\eta \geq 0.70$: exploratory.
- $\eta = 0.30$ also gives $\chi_1^2 = 4.05$ ($p = 0.044$): one flag among twelve tests at the 5% level is expected once. Reported, not tuned.
- Claim range for the paper: $\eta \leq 0.65$ **validated**, 0.67–0.69 “no bulk value”, ≥ 0.70 exploratory.

Exercise 1

Z at four sizes: 2.075, 2.068, 2.066, 2.065 for $N = 100, 400, 900, 1600$. Write $x = 1/\sqrt{N}$ (0.1, 0.05, 0.0333, 0.025). Fit a line by eye (or with two points, $N = 100$ and 1600): what is Z_∞ , what is the slope a ?

2.2 Speed of sound, route A: the instrument works

In plain words

We do not measure sound directly. We put a movable wall (the divider) between two gas columns and let it swing. The gas on both sides acts like a spring whose stiffness is set by the sound speed. Measure the swing frequency ν for 9 divider masses; theory (the Román relation) says all 9 points lie on one straight line through the origin whose slope is c_s . Nine masses give nine independent checks of the same number.

$$\nu = c_s \frac{K}{2\pi L_{\text{eff}}}, \quad \cot K = \frac{M}{2Nm} K, \quad L_{\text{eff}} = L_0 - 2r. \quad (2)$$

Why a fit “through the origin”

The relation has no constant term: a divider that cannot swing ($K \rightarrow 0$) has frequency zero. So we fit $\nu = c_s x$ with $x = K/(2\pi L_{\text{eff}})$. A second fit with a free intercept b is the alarm: if b is not zero within error, the single-swing-mode picture is being stretched (at $\eta = 0.57\text{--}0.63$ the compartment is only 6–7 diameters long and b absorbs up to 3.6%).

Why route A validates the instrument but not the bulk fluid

Route A always has 50 disks per side. To change η it changes the box length, $L_0 = 3.93 \sigma/\eta$: 196 diameters at $\eta = 0.02$, 6 at $\eta = 0.65$. So density and confinement change together and a deviation from KR could be either. That is fine for showing the method works (the dilute end reproduces $\sqrt{2}(1 + 2\eta)$ with the right slope), but a bulk claim needs the $1/\sqrt{N}$ game from the pressure section: several N at the *same* η and the same box shape.

Numbers to write down

- Refit (through the origin, 7186 of 7200 runs eligible): dilute end $-0.49 \dots -0.02\%$ vs KR; $\eta = 0.11\text{--}0.39$: $+0.5 \dots +1.1\%$; $\eta = 0.52\text{--}0.55$: $+2 \dots 2.5\%$; $\eta \geq 0.65$: not a fluid-branch value (structure in a 6-diameter box).
- Dilute slope: $c_s/\sqrt{2} - 1 = 0.035$ at $\eta = 0.0196$ vs theory $2\eta = 0.039$. The ideal-gas limit is right.
- Per-mass residuals at the dilute end scatter by $0.5\text{--}0.7\%$ with no trend in M and no growth with η : instrument scatter, not damping, not the L_{eff} convention. Closed.
- The 504 `wall_overdue` flags: two disks seeded exactly on the right wall (float rounding at 9800 px); they reflect once at $t = 0$; harmless; runs kept. The seeder pad must be fixed (double precision) before any *new* run at $L_0 \geq 100$.

Exercise 2

$\eta = 0.05$, $N = 50$ per side, $r = 0.5$, $H = 10$. Compute L_0 , L_{eff} , the mean free path $\lambda = 0.555/\eta$, and $\text{Kn} = \lambda/L_0$. Then take $M = 100$ ($\alpha = 1$, $K = 0.860$) and $c_s = 1.567$: what is ν and the swing period?

2.3 The gap: no fixed-geometry c_s data below $\eta = 0.67$

In plain words

We do have a family with $N = 100, 400, 900, 1600$ at the same box shape (“famA”), but it was run for the dense side, $\eta = 0.670\text{--}0.730$. At 0.670 the four sizes sit perfectly on a $1/\sqrt{N}$ line ($\chi^2_2 = 0.07$) and give $c_\infty = 10.65 \pm 0.04$, which is 6.4% below KR — the same message as pressure: at 0.67 the fluid EOS is already off. Everything above 0.67 is not even monotone in N (χ^2 from 36 to 723): the coexistence and hexatic region, no bulk value, as expected.

So for Paper 1 we currently have a *bulk* pressure claim and only an *instrument* sound-speed claim. The missing piece is a small family at $\eta = 0.10, 0.30, 0.50, 0.60, 0.65$ (“famB”). It is the last c_s campaign of Paper 1.

Numbers to write down

famA at $\eta = 0.670$: $c_s = 13.04$ ($N=100$), 11.86 (400), 11.48 (900), 11.25 (1600) $\rightarrow c_\infty = 10.65 \pm 0.04$, slope $a = +23.9$. Finite-size effects at $N = 100$ are $+22\%$ here — which is why route A alone cannot be the bulk claim.

Exercise 3

Using the famA numbers above, check c_∞ with the two-point trick ($N = 100$ and 1600 , $x = 0.1$ and 0.025). How far is your answer from 10.65 ? Why is the 4-point weighted fit better?

2.4 Level 0: the old energy-transfer runs cannot be used

In plain words

Level 0 is bookkeeping: the work the piston does must show up somewhere — as gas kinetic energy, divider kinetic energy, spring energy, or heat into a thermal wall. With elastic walls there is no heat, so

$$W = \Delta KE_{\text{gas}} + \Delta KE_{\text{div}} + \Delta E_{\text{spring}} \quad \text{exactly, up to round-off.} \quad (3)$$

Claude Code looked at the existing runs (`mass_sweep_eta02_N600`) and found two show-stoppers. First, they were run with `--edmd-acc=1`, the accelerated backend that failed validation in August (181 of 450 trajectories invalid). Second, the trace never recorded the gas kinetic energy, nor per-collision impulses, nor gas momentum. So one term of the ledger is simply missing: the residual is *undefined*, not small. Nothing from those runs is physics; they told us what to log. The good news: the collision rule in the code is exactly the textbook one, verified line by line. And the old “efficiency” number `SpringE_max/W_in` (peak spring energy over piston work) is not a thermodynamic quantity — a single-instant maximum of an oscillating store — so we stop using it.

Numbers to write down

Piston rule, prescribed-velocity piston with velocity u , particle normal velocity v before the hit:

$$v' = 2u - v, \quad \Delta E = 2mu(u - v), \quad \Delta p = 2m(u - v). \quad (4)$$

Existing runs: $N = 600$, $\eta = 0.10$, work target $2 \times 10^4 k_B T$, piston travel 10σ in about one σ -time (u/c_s of order 2–3: a shock).

Exercise 4

Piston moves into the gas at $u = 0.05$. A particle comes toward it with $v = -1.0$. Compute v' , ΔE , Δp . Now a particle moving away from the piston at $v = +0.03$ (slower than the piston): is it hit, and what is ΔE ? And $v = +0.10$: is it hit at all?

2.5 Running now: $N = 2500$ pressure at $\eta = 0.65, 0.67, 0.69$

In plain words

A fifth box size, to see whether the $1/\sqrt{N}$ line at 0.65 holds with a further point (it should, within the quoted error) and whether 0.67 stays off the line (it should). Analysis-only until it lands; then table B is recomputed with 5 sizes (3 degrees of freedom).

What a wrong result looks like

Z_∞ at 0.65 shifting by more than 0.2%, or χ^2 at 0.65 jumping above ≈ 6 , would mean the $1/\sqrt{N}$ form is not the right finite-size law there — then the claim range moves down to 0.60. A calibration row with a large standard error (now logged) is the first thing to look at if that happens.

3 The plan, step by step

3.1 Step 1 — driver logging (no physics change)

In plain words

Before any new push run, the trace must record gas kinetic energy (total, left, right) and the gas momentum $\sum mv_x$ per sample, and a switchable (“gated”) per-event file must record every piston and divider collision: time, piston speed u , particle speed before and after, ΔE , Δp . “Gated” means: off unless an environment variable turns it on, so accepted runs are bit-for-bit unchanged. The acceptance test is literal: rerun one accepted sound-speed cell and `cmp` the divider trace against the accepted file — zero differing bytes.

What a wrong result looks like

Any byte difference in the accepted trace, or a change in any mode other than `energy_transfer`, and the change is rejected. Also: units. The velocity columns must be declared as σ/σ -time; a factor 24 hiding in one column would make ΔKE_{div} wrong by 576.

3.2 Step 2 — Level 0 pilot: one slow push, 50 disks per side

In plain words

Smallest experiment that can close the ledger: $\eta = 0.10$, $L_0 = 39.3$, elastic walls, validated core, piston at $u = 0.05$ (about 3% of the sound speed) compressing the right compartment by 10%, then a hold of 200 σ -time so the divider rings down. Five seeds. From it we get four checks at once.

Numbers to write down

1. Ledger residual $R(t) = W - [\Delta KE_{\text{gas}} + \Delta KE_{\text{div}} + \Delta E_{\text{spring}}]$: expect $|R| \sim 10^{-10}$ – $10^{-8} k_B T$ (double-precision round-off over $\sim 10^4$ collisions).
2. Sum of per-event $\Delta E = \text{logged PistonWork}$ to the last digit.
3. Momentum: $\Delta(p_{\text{gas}} + p_{\text{div}}) = \sum \Delta p$ (pistons + outer walls) to round-off.
4. Adiabatic prediction for the compressed side ($N = 50$, $\eta : 0.10 \rightarrow 0.11$):

$$\frac{T_f}{T_i} = \exp \int_{0.10}^{0.11} \frac{Z(\eta)}{\eta} d\eta = 1.127 \text{ (KR)}, \quad \text{ideal gas } 1.100, \quad (5)$$

$$W_{\text{qs}} = N k_B T_i \left(\frac{T_f}{T_i} - 1 \right) = 6.3 k_B T. \quad (6)$$

5. Dissipation $W_{\text{diss}} = \langle W \rangle - W_{\text{qs}}$: the collisionless estimate $W_{\text{diss}}/W_{\text{qs}} \approx 4.5 u/c_s = 4.5 \times 0.029 = 13\%$ is an upper-side number; in the fluid expect roughly 0.3 – $0.8 k_B T$ on top of $6.3 k_B T$.

What a wrong result looks like

$|R| > 10^{-6} k_B T$: a term is missing or double-counted (check whether the divider’s own work is booked on the gas side or the divider side). $W_{\text{diss}} < 0$ beyond the seed error: W_{qs} computed with the wrong η range or wrong N . W_{diss} of tens of $k_B T$: units, or the push was much faster than specified.

Exercise 5

Ideal-gas version of item 4 by hand: $N = 50$, $k_B T_i = 1$, area shrinks by 10% ($A_f = 0.909 A_i$). $T_f/T_i = A_i/A_f = ?$, $W_{qs} = ?$ Why is the KR answer (6.3) larger than the ideal one?

3.3 Step 3 — famB: the bulk sound-speed claim

In plain words

$\eta = 0.10, 0.30, 0.50, 0.60, 0.65$; $N = 100, 400, 900, 1600$ at the famA box shape; 5 masses; 10 repeats. Same manifest treatment, through-origin fits, $1/\sqrt{N}$ extrapolation with χ^2 . Result: a table $c_\infty(\eta)$ vs KR that sits next to the pressure table. Launched only after $N = 2500$ finishes (machine time).

Numbers to write down

Expected from route A and the EOS: c_∞ within $\approx 1\%$ of KR at 0.10–0.50 ($c_s = 1.744, 2.852, 5.416$), and a visible finite-size slope that grows with η (famA showed +22% at $N = 100$ for $\eta = 0.67$).

3.4 Step 4 — Level 1: the divider as a thermometer and a friction meter

In plain words

Hold the piston still, let the divider jiggle in equilibrium. Two things come out. Equipartition: the divider is one more degree of freedom, so $\frac{1}{2}M \langle \dot{\xi}^2 \rangle = \frac{1}{2}k_B T$ — a temperature check that costs nothing. And the force on a *held* divider fluctuates; the time-integral of its autocorrelation is the friction ζ that a *moving* divider will feel:

$$\zeta = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt, \quad W_{\text{diss}} \simeq \int \zeta \dot{\lambda}^2 dt. \quad (7)$$

This is the “predict dissipation from equilibrium” trick (Sivak–Crooks); Step 6 checks it.

3.5 Step 5 — Level 2: many pushes, the work histogram, Jarzynski

In plain words

Repeat the push 10^3 – 10^4 times from independent equilibrium starts (Maxwellian velocities at T , equilibrated positions — “canonical initial sampling”; no thermal wall is needed during the push). Every run gives one number W . The histogram $P(W)$ has a slow-push limit (a spike at W_{qs}) and a fast-push limit (Lua–Grosberg: each particle hit at most once, $P(W)$ follows from the Maxwell distribution alone). In between, Jarzynski says

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \Delta F = N k_B T \int_{\eta_i}^{\eta_f} \frac{Z}{\eta} d\eta \quad (= 6.0 k_B T \text{ for 50 disks, } 0.10 \rightarrow 0.11). \quad (8)$$

Why the piston must be slow for this

The average of $e^{-\beta W}$ is dominated by the rare runs with unusually small W . You need about $e^{\beta W_{\text{diss}}}$ runs to catch them: $\beta W_{\text{diss}} = 3 \rightarrow 20$ runs, $5 \rightarrow 150$, $10 \rightarrow 2 \times 10^4$. So keep W_{diss} below ≈ 5 – $8 k_B T$: small N , $u/c_s \leq 0.3$, compression 10–25%. The old shock runs had βW_{diss} in the thousands — hopeless for this.

Exercise 6

You measure βW in five runs: 6.5, 6.9, 6.2, 7.4, 6.6 with $\beta \Delta F = 6.0$. Compute $\langle W \rangle$, W_{diss} (with $W_{\text{qs}} = 6.3$), and the Jarzynski estimate $-\ln \frac{1}{5} \sum e^{-\beta W_k}$. Is $\langle W \rangle \geq \Delta F$? Is the Jarzynski estimate closer to ΔF than $\langle W \rangle$ is?

3.6 Step 6 — Level 3: dissipation vs speed

In plain words

Same push at $u/c_s = 0.01, 0.03, 0.1, 0.3$. Plot W_{diss} against u . Linear response predicts a straight line through the origin at small u with slope $\zeta \Delta x$ from Step 4. Where the line bends, linear response ends — that is a result, not a failure.

3.7 Step 7 — Level 4: memory, prediction, nostalgia

Mutual information with a coin

Two coins X and Y . If they are independent, knowing X tells you nothing about Y : $I(X; Y) = 0$. If they are glued together ($Y = X$), knowing X tells you everything: $I = H(X) = \ln 2$ nats = 1 bit. In general

$$I(X; Y) = \sum_{x,y} p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} \geq 0. \quad (9)$$

The three quantities

Let x_t be the piston's command at step t (drawn at random from a rule, so that it carries information) and s_t the state of gas + divider (coarse-grained, e.g. divider position and velocity in bins).

$$I_{\text{mem}} = I(s_t; x_t), \quad I_{\text{pred}} = I(s_t; x_{t+1}), \quad I_{\text{nost}} = I_{\text{mem}} - I_{\text{pred}} \geq 0. \quad (10)$$

Memory: what the system's state tells you about the command *now*. Prediction: what it tells you about the *next* command. Nostalgia: memory that is useless for the future. It cannot be negative, because the state learns about the future only through the present (data processing: $s_t \rightarrow x_t \rightarrow x_{t+1}$). The inequality in your screenshot, $H(Y) \leq H(Z)$ for $Y = f(Z)$, is the one-variable version: coarse-graining never creates information.

The bound we want to show

$$\beta \langle W_{\text{diss}}[x_t \rightarrow x_{t+1}] \rangle \geq I_{\text{nost}}. \quad (11)$$

Left side: from the closed ledger (Steps 1–2). Right side: from histograms of (s_t, x_t) and (s_t, x_{t+1}) over the trajectories. Every nat of useless memory costs at least $k_B T$ of dissipation. For the bound to be a real test both sides must be of order one: that is why N is small and $u/c_s \leq 0.3$.

Numbers to write down

Plug-in estimator bias $\approx (k_s - 1)(k_x - 1)/(2N_{\text{traj}})$ nats. $k_s = 10$ state bins, $k_x = 3$ commands: 0.009 nats at 10^3 trajectories, 0.001 at 10^4 . I_{nost} must sit clearly above that.

Exercise 7

Joint table of a command $x \in \{\text{push, hold}\}$ and a state bin $s \in \{\text{left, right}\}$ over 100 trajectories: push&left 40, push&right 10, hold&left 10, hold&right 40. Compute $p(x)$, $p(s)$, and $I(s; x)$ in nats. (Sanity: independent would give 25 in every cell and $I = 0$.)

4 What to write on the first page of your notes

1. Paper 1 claim: Z bulk-validated $\eta \leq 0.65$; c_s instrument-validated on route A; c_s bulk claim pending famB.
2. Level 0 closes after: logging change $\rightarrow$ cmp test $\rightarrow$ 5-seed pilot with $|R| \sim$ round-off, $W_{\text{qs}} = 6.3 k_B T$, $W_{\text{diss}} \approx 0.3\text{--}0.8 k_B T$.
3. Design window for Paper 2: $N = 50\text{--}300$ per side, $u/c_s = 0.01\text{--}0.3$, compression 10–25%, $\beta W_{\text{diss}} \lesssim 5$, $10^3\text{--}10^4$ trajectories, stochastic piston command.
4. Two positive controls waiting in the queue: route A to $\eta = 0.01, 0.005$ (after the seed-pad fix) and the Knudsen pilot $N_{\text{side}} = 25, 10$ at $L_0 = 200$ (expect c_s to rise from $\sqrt{2}$ toward $\sqrt{3}$).

5 Answers

1. Two points: slope $a = (2.075 - 2.065)/(0.1 - 0.025) = 0.133$; $Z_\infty = 2.065 - 0.133 \times 0.025 = 2.062$. (A weighted 4-point fit gives about the same; the message is that $N = 100$ is 0.6% high and the infinite-box value is below every measured point.)
2. $L_0 = 3.927/0.05 = 78.5$; $L_{\text{eff}} = 77.5$; $\lambda = 11.1$; $\text{Kn} = 0.141$. $\nu = 1.567 \times 0.860/(2\pi \times 77.5) = 2.77 \times 10^{-3}$ per σ -time; period 361 σ -time.
3. $a = (13.04 - 11.25)/(0.1 - 0.025) = 23.9$; $c_\infty = 11.25 - 23.9 \times 0.025 = 10.65$. Same as the full fit here because the two end points dominate; the 4-point weighted fit is better because it uses the error bars and gives χ^2 , which tells you whether a line is even the right model.
4. $v = -1.0$: $v' = 2(0.05) + 1.0 = 1.10$; $\Delta E = 2(0.05)(0.05 + 1.0) = 0.105$; $\Delta p = 2(1.05) = 2.10$. $v = +0.03$: the piston (0.05) catches it; $v' = 0.07$; $\Delta E = 2(0.05)(0.02) = 0.002$. $v = +0.10$: it outruns the piston, no hit ($\Delta E = 0$).
5. $T_f/T_i = 1/0.909 = 1.100$; $W_{\text{qs}} = 50 \times 0.100 = 5.0 k_B T$. KR gives more because $Z > 1$: the gas pushes back harder than an ideal gas, so compressing it by the same amount costs more work ($Z \approx 1.24$ at $\eta = 0.1$).
6. $\langle \beta W \rangle = 6.72$; $W_{\text{diss}} = 6.72 - 6.3 = 0.42 k_B T$. The five values of $e^{-\beta W}$ are 1.50, 1.01, 2.03, 0.61, 1.36 (all $\times 10^{-3}$); their mean is 1.30×10^{-3} and $-\ln(1.30 \times 10^{-3}) = 6.64$. Yes, $\langle W \rangle \geq \Delta F$. The Jarzynski estimate (6.64) is closer to 6.0 than $\langle W \rangle$ (6.72) is, but still high: with only five samples the estimate is dominated by the single smallest W (the 6.2 run) and its statistical error is large; the rule $e^{\beta W_{\text{diss}}} \approx e^{0.7} \approx 2$ says five runs are enough to see the effect, not to pin ΔF to $0.1 k_B T$. With 10^3 runs it converges.
7. $p(\text{push}) = p(\text{hold}) = 0.5$, $p(\text{left}) = p(\text{right}) = 0.5$. $I = 2 \times 0.4 \ln \frac{0.4}{0.25} + 2 \times 0.1 \ln \frac{0.1}{0.25} = 0.8 \times 0.470 + 0.2 \times (-0.916) = 0.376 - 0.183 = 0.193$ nats (0.28 bits).